Moumita Sen Sarma

PhD Candidate in Computer Science · Explainable AI · Computer Vision · Knowledge Graphs & LLM/RAG
Manhattan, KS, USA · Open to relocation · +1 (785) 317-4853 · moumita@ksu.edu
moumita-sen-sarma.github.io · github.com/Moumita-Sen-Sarma · linkedin.com/in/moumita-sen-sarma-91245413a · https://scholar.google.com/citations?user=7uRtCK0AAAAJ

Summary

I am a PhD candidate in Computer Science at Kansas State University. My research focuses on explainable and neurosymbolic AI, computer vision, and knowledge graphs. I have 9 publications in peer-reviewed conferences and journals. I also develop applied AI systems using knowledge graphs and retrieval-augmented generation.

Education

Kansas State University, Manhattan, KS, USA — 08/2024 – Expected Spring 2029
PhD in Computer Science, PhD Candidate (All But Dissertation)

Chittagong University of Engineering & Technology (CUET), Bangladesh — 2016 – 2021
B.Sc. in Computer Science & Engineering. CGPA: 3.89/4.00; Merit Position: 1st of 120

Research and Work Experience

AI Consultant, K-State AI Consultation Services (KAICS), Kansas State University (June 2026 - Present)

- Provide AI consultation to faculty, staff, and graduate students on research projects.
- Developed and delivered tutorials on Fine-Tuning LLMs, Agentic AI, and Image Processing with AI (approx. 25 participants each).

Graduate Research Assistant, Data Semantics Lab, Kansas State University (Aug 2024 - May 2026)

- Developed a knowledge graph for wheat fungicide management (GRIP project), covering fungicide properties, disease observations, and yield outcomes. Deployed it on GraphDB with SPARQL querying and built a Flask-based query and visualization interface.

Lecturer, Department of Computer Science & Engineering, CUET, Bangladesh (Jul 2022 - Jul 2024)

- Taught Software Engineering, Computer Graphics, Programming for Planners, and Electrical Devices and Circuits.
- Organizing committee member, ECCE 2023 (international conference) and CLBLP 2023 (workshop).

Lecturer, Department of Computer Science & Engineering, East Delta University, Bangladesh (Sept 2021 - Jul 2022)

- Taught Digital Image Processing, Pattern Recognition, Computer Graphics, Compiler Design, OOP, and Operating Systems.

Selected Research and Projects

ThreatGraph-RAG: Hybrid GraphRAG for Cyber Threat Intelligence — github.com/Moumita-Sen-Sarma/ThreatGraph-RAG · threatgraph-rag.streamlit.app

- Developed a question-answering system combining Neo4j graph retrieval, Qdrant vector search, and OpenAI GPT-5.6.
- Built the knowledge graph from MITRE ATT&CK and CISA KEV data.
- Implemented entity resolution, citation validation, and guardrails for grounded answers.
- Exposed the system via a Model Context Protocol (MCP) server and deployed it on Streamlit.

Hidden-Layer Analysis of Real vs. AI-Generated Images in CNNs (first author, NeSy 2026)

- Analyzed CNN hidden-layer representations of real and AI-generated images.

Neuron-Level Interpretability with Concept Induction (K-CAP 2025 poster; Elsevier book chapter)

- Applied concept induction (ECII) to explain hidden CNN neurons with human-understandable concepts.

AVaTER: Multimodal Emotion Recognition (Sensors 2024)

- Co-developed a cross-modal attention model fusing audio, visual, and text for emotion recognition.

Traditional Bangladeshi Sports Video Classification (first author, Applied Sciences 2021) — github.com/Moumita-Sen-Sarma

- Developed a CNN-LSTM model to classify 5 traditional sports from video, with a Flask web interface.

Publications

- **M. S. Sarma**, P. Hitzler, E. Y. Vasserman. "Do CNNs Internally Represent Real and Fake Images Differently? A Hidden-Layer Analysis." *20th Conference on Neurosymbolic Learning and Reasoning (NeSy 2026)*. Preprint on arXiv.
- A. Dalal, **M. S. Sarma**, A. Das, S. E. Akkamahadevi, E. Y. Vasserman, P. Hitzler. "Neurosymbolic Hidden Neuron Analysis in Convolutional Neural Networks." In *Neuro-Symbolic AI: Bridging the Gap Between Neural Networks and Symbolic Reasoning*, Elsevier, 2025 (book chapter, accepted, in press).
- A. Barua, A. Das, **M. S. Sarma**, P. Hitzler. "Symbolic Reasoning Methods in Neural Architectures and Large Language Models." In *Neuro-Symbolic AI: Bridging the Gap Between Neural Networks and Symbolic Reasoning*, Elsevier, 2025 (book chapter, accepted, in press).

- A. Das, **M. S. Sarma**, M. M. Hoque, N. Siddique, M. A. A. Dewan. "AVaTER: Fusing Audio, Visual, and Textual Modalities Using Cross-Modal Attention for Emotion Recognition." *Sensors*, 24, 5862, 2024.
- S. S. Aonty, K. Deb, **M. S. Sarma**, P. K. Dhar, T. Shimamura. "Multi-Person Pose Estimation Using Group-Based Convolutional Neural Network Model." *IEEE Access*, 11, 42343-42360, 2023.
- S. S. Aonty, **M. S. Sarma**, K. Deb. "Jointly Learning Structure for Human Pose Estimation using Convolutional Neural Networks." *ICECTE 2022*, IEEE.
- **M. S. Sarma**, K. Deb, P. K. Dhar, T. Koshiba. "Traditional Bangladeshi Sports Video Classification Using Deep Learning Method." *Applied Sciences*, 11(5), 2149, 2021.
- **M. S. Sarma**, A. Das. "BMGC: A Deep Learning Approach to Classify Bangla Music Genres." *NISS 2021*, ACM.
- A. M. Uddin, A. Al Miraj, **M. S. Sarma**, A. Das, M. M. Gani. "Traditional Bengali Food Classification Using Convolutional Neural Network." *IEEE TENSYMP 2021*.

Posters:

- **M. S. Sarma**, S. E. Akkamahadevi, P. Hitzler. "A Case Study on Concept Induction for Neuron-Level Interpretability in CNN." *K-CAP 2025*. View Poster
- **M. S. Sarma**, A. Gautam, N. Gelal, et al. "Building Knowledge Graph for Fungicide Management in Wheat." Kansas State University, 2025. View Poster

Technical Skills

- **Programming:** Python, C/C++, Java, SQL
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, CNN, LSTM, Transformers, Hugging Face, computer vision, multimodal learning, video classification, pose estimation
- **LLMs & Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), GraphRAG, OpenAI API, Model Context Protocol (MCP), Sentence Transformers, embeddings, semantic search, diffusion models (SDXL, FLUX)
- **Explainable & Neurosymbolic AI:** Explainable AI (XAI), model interpretability, concept induction (ECII), neurosymbolic AI
- **Knowledge Graphs & Semantic Web:** Neo4j, GraphDB, RDF, OWL, SPARQL, Protégé, ontology engineering, entity resolution
- **Databases:** Qdrant (vector database), MySQL
- **Tools & MLOps:** Docker, Kubernetes, FastAPI, Flask, Streamlit, Git

Honors and Leadership

- Graduate Student of the Month, Carl R. Ice College of Engineering, Kansas State University (09/2025)
- Vice President, Computer Science Graduate Student Association, Kansas State University (2026)
- Chairperson, IEEE CUET WIE Affinity Group Student Branch (2019–2020)

Certifications

- Machine Learning in Production - DeepLearning.AI (Coursera)
- Build Basic Generative Adversarial Networks (GANs) - Coursera
- Convolutional Neural Networks in TensorFlow - Coursera
- Machine Learning - Coursera
- Docker for Absolute Beginners; Kubernetes for Absolute Beginners – KodeKloud

References

Dr. Pascal Hitzler

University Distinguished Professor and Lloyd T. Smith Creativity in Engineering Chair
Director, Center for Artificial Intelligence and Data Science (CAIDS)
Department of Computer Science, Kansas State University, Manhattan, KS 66506, USA
Email: hitzler@k-state.edu Phone: +1 (785) 532-7951
Relationship: PhD Major Supervisor (2024 - Present)

Dr. Eugene Vasserman

Associate Professor
Director, Center for Cybersecurity and Trustworthy Systems
Department of Computer Science, Kansas State University, Manhattan, KS 66506, USA
Email: eyv@k-state.edu
Relationship: Research Collaborator and Co-author (NeSy 2026; Elsevier book chapter)